

James Tour and his colleagues have developed gold nanomesh electrodes that offer ultra-high stretchability. At the same time, they can provide good electrical conductivity. According to Tour and Co, the current crop of materials fall short as reliable super-capacitors, and their new technology is well on its way to become the viable alternative. "While flexible gadgets such as 'electronic skin' and roll-up touchscreens are moving ever closer to reality, their would-be power sources are either too wimpy or too stiff. But that's changing fast," the researchers were quoted in the *Journal of the American Chemical Society* (ACS).

Tour's 'flexible' material consists of nanoporous nickel-fluoride electrodes layered around a solid electrolyte providing battery-like super-capacitor performance. These nano-pores allow ions to flow easily throughout the material, so much so that the resulting structure can pack in far more power for its size. The material is thus able to provide the best qualities of a high-efficiency battery as well as a high-powered super-capacitor. Researchers claim that you can bend and fold the film and recharge it thousands of times with little or no loss in performance.

Have some lettuce... grown in Fujitsu's Fab!

Are you a food connoisseur? Even if you are not, how would you like to taste a patch of iceberg lettuce grown in a hi-tech clean room powered by Internet of Things smartness?

Semiconductor-firm Fujitsu began selling its first round of 'Kirei Yasai' lettuce on May 7th this year. The salad ingredient is grown in a 2000 square metre facility that previously used to fabricate microchips and other components for electronics products.

These lettuce are the result of a lot of sensor-based arrays and systems working in tandem with cloud computing solutions to crunch data and ensure that the environment remains at the optimal levels throughout the growth process. This includes monitoring temperature, humidity and

fertiliser composition as they try to figure out the best growing conditions and ways to control micro organisms.

The project uses 'Akisai,' the food and agricultural cloud from Fujitsu that leverages ICT to dramatically improve the efficiency of agricultural operations. Akisai will be used to continually analyse agricultural data for highly productive cultivation and to facilitate the entire management process, including management, production and sales, for a more efficient agricultural operation.

Are birds getting lost because of our electronics?

This debate is over. And the answer is 'Yes.' A letter published online in the

journal 'Nature' goes to prove that migratory birds are unable to use their magnetic compass in the presence of man-made electromagnetic noise around them.

The test was conducted on European Robins, by exposing them to electromagnetic noise present in unscreened wooden huts at the University of Oldenburg campus. Upon being exposed, these birds were unable to use their internal magnetic compass and lost orientation. When aluminium-screened huts were used, which attenuated the electromagnetic noise, their capability to orient themselves to the magnetic field reappeared.

The question now is: Does it affect biological processes and human health as strongly as it affects these poor birds?

Futuristic mind-controlled prosthetic arm is here

DARPA has finally succeeded in getting FDA approval for their prosthetic arm with near-natural control. Named the DEKA arm system, it is an advanced electromechanical prosthetic arm that allows almost full control of the arm via electrical signals from electromyogram electrodes.

It allows for simultaneous control of multiple joints using a variety of input devices, including wireless signals generated by innovative sensors on the user's feet. It can handle anything from sensitive items like grapes and eggs, all the way to power tools like a hand drill.

M2M technology to help turtles and tourism

Telit Wireless Solutions, a global provider of high-quality machine-to-machine (M2M) solutions, products and services, has announced that Turtle Sense, a project designed to accurately predict when sea turtles hatch and emerge from their nests, can integrate Telit HE910 modules to communicate sensor data. The goal



Telit HE910 modules to communicate sensor data

is to protect both the threatened and endangered sea turtles and the area's critical tourism industry. The Turtle Sense technology was developed by Nerds-Without-Borders.

Because coastal economies rely heavily on tourism from public access to beaches, business and environmental interests must balance the popularity of coastal towns with requirements for the preservation of sea turtle nesting areas prescribed in The Endangered Species Act. Currently, there is no reliable way to predict when the tiny turtles will emerge from their nests near the dune line and parade to the surf.

In order to extend battery life throughout the nesting, incubation and hatching period, transmission timing starts slow and intermittent and increases in frequency once activity is reported.

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